



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ASSIGNMENT 9

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1. Write a C++ program to write number 1 to 200 in a data file NUM.TXT.

```
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      ofstream fout("NUM.TXT");
7
8      for (int i = 1; i <= 200; i++) {
9          fout << i << " ";
10     }
11
12     fout.close();
13     return 0;
14 }
```

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61
62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90
91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114
115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136
137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158
159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180
181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200
```

2. Write a user-defined function in C++ to read NOTES.TXT and count alphabets.

```
1  #include <iostream>
2  #include <fstream>
3  #include <cctype>
4  using namespace std;
5
6  void countAlphabets() {
7      ifstream fin("NOTES.TXT");
8      char ch;
9      int count = 0;
10
11     while (fin.get(ch)) {
12         if (isalpha(ch)) {
13             count++;
14         }
15     }
16
17     cout << "Number of alphabets: " << count;
18     fin.close();
19 }
20
21 int main() {
22     countAlphabets();
23     return 0;
24 }
```

3. Write a program to copy contents of one file to another.

```
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      ifstream fin("source.txt");
7      ofstream fout("destination.txt");
8      char ch;
9
10     while (fin.get(ch)) {
11         fout.put(ch);
12     }
13
14     fin.close();
15     fout.close();
16     return 0;
17 }
```

4. Program for string I/O operations with file handling

```
1  #include <iostream>
2  #include <fstream>
3  #include <cstring>
4  using namespace std;
5
6  int main() {
7      char str[100];
8
9      cout << "Enter a string: ";
10     cin.getline(str, 100);
11
12     // Length calculation
13     cout << "Length: " << strlen(str) << endl;
14
15     // Store in file
16     ofstream fout("data.txt");
17     fout << str;
18     fout.close();
19
20     // Read from file
21     ifstream fin("data.txt");
22     char ch;
23
24     cout << "Stored string: ";
25     while (fin.get(ch)) {
26         cout << ch;
27     }
28
29     fin.close();
30     return 0;
31 }
```

5) (a) Create A-Z and read 10th character using seekg()

```
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      ofstream fout("AZ.txt");
7
8      for (char ch = 'A'; ch <= 'Z'; ch++) {
9          fout << ch;
10     }
11     fout.close();
12
13     ifstream fin("AZ.txt");
14     fin.seekg(9); // 10th character (index 9)
15
16     char ch;
17     fin.get(ch);
18     cout << "10th character: " << ch;
19
20     fin.close();
21     return 0;
22 }
```

(b) Overwrite 5th character using seekp()

```
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      fstream file("AZ.txt", ios::in | ios::out);
7
8      file.seekp(4); // 5th position
9      file.put('Z');
10
11     file.close();
12     return 0;
13 }
```

(c) Find file size using tellg()

```
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      ifstream fin("AZ.txt", ios::binary | ios::ate);
7
8      cout << "File size: " << fin.tellg() << " bytes";
9
10     fin.close();
11     return 0;
12 }
```

(d) Read last character using seekg()

```
C: > Users > Aadhya > OneDrive > Desktop > C++ > .vscode > DSA > .vscode > files.cpp
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      ifstream fin("AZ.txt");
7
8      fin.seekg(-1, ios::end);
9
10     char ch;
11     fin.get(ch);
12
13     cout << "Last character: " << ch;
14
15     fin.close();
16     return 0;
17 }
```

```
PS C:\Users\Aadhya\OneDrive\Desktop\C++\.vscode\DSA\.vscode> cd "c:\Users\Aadhya\OneDrive\Desktop\C++\.vscode\DSA\.vscode\" ; if ($?) { g++ files.cpp -o files }
; if ($?) { .\files }
File size: -1 bytes
```

(e) data.txt operations

```
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      ofstream fout("data.txt");
7      fout << "This is a sample file\nWith multiple lines\nFor testing";
8      fout.close();
9
10     ifstream fin("data.txt");
11
12     fin.seekg(10);
13     cout << "Current position: " << fin.tellg() << endl;
14
15     char ch;
16     cout << "Remaining content:\n";
17     while (fin.get(ch)) {
18         cout << ch;
19     }
20
21     fin.close();
22     return 0;
23 }
```

```
PS C:\Users\Aadhya> cd "c:\Users\Aadhya\OneDrive\Desktop\C++\.vscode\DSA\.vscode\" ; if ($?) { g++ files.cpp -o files } ; if ($?) { .\files }
Remaining content:
sample file
With multiple lines
For testing
PS C:\Users\Aadhya\OneDrive\Desktop\C++\.vscode\DSA\.vscode>
```

6. Hello World file operations

```
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4
5  int main() {
6      fstream file("hello.txt", ios::in | ios::out | ios::trunc);
7
8      string text = "Hello World";
9
10     // Write and show tellp()
11     for (int i = 0; i < text.length(); i++) {
12         file.put(text[i]);
13         cout << "Position after writing '" << text[i] << "': " << file.tellp() << endl;
14     }
15
16     // Replace "World" with "C++"
17     file.seekp(6); // Position of 'W'
18     file << "C++";
19
20     file.close();
21
22     // Display updated content
23     ifstream fin("hello.txt");
24     char ch;
25
26     cout << "\nUpdated content: ";
27     while (fin.get(ch)) {
28         cout << ch;
29     }
30
31     fin.close();
32     return 0;
33 }
```
